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CASE REPORT *Reproductive genetics*

PGD for the m.14487 T>C mitochondrial DNA mutation resulted in the birth of a healthy boy

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ABSTRACT: We report on the first PGD performed for the m.14487 T>C mitochondrial DNA (mtDNA) mutation in the *MT-ND6* gene, associated with Leigh syndrome. The female carrier gave birth to a healthy baby boy at age 42. This case adds to the successes of PGD for mtDNA mutations.

Key words: PGD / mitochondrial DNA (mtDNA) mutation / T14487C / MT-ND6 / Leigh syndrome

Introduction

Mitochondrial DNA (mtDNA) is subject to exclusive maternal inheritance. Pathogenic mutations in the mtDNA are implicated in a variety of (mostly) multisystem disorders with variable organ involvement, age of onset and severity, generally without treatment options.

Leigh syndrome, a progressive neurodegenerative disorder with multisystem involvement which is often lethal in childhood, is an example at the severe end of the mitochondrial disease spectrum (Gerards *et al.*, 2016). Leigh syndrome can be caused by mutations in a variety of genes, either originating from the nuclear DNA or from the mtDNA (Gerards *et al.*, 2016). For the former, transmission to offspring and recurrence risks follow Mendelian patterns, mostly of autosomal recessive inheritance. The applicability of reproductive options, i.e. prenatal diagnosis (PND) and PGD, is similar to that of other recessive disorders. However, when caused by an mtDNA mutation, Leigh patients' carrier mothers encounter challenges in pursuing unaffected offspring.

Due to the heteroplasmic nature of mtDNA mutations, referring to the coexistence of mutant and normal mtDNA in the carrier female's cells, and its complex transmission behavior (Johnston *et al.*, 2015), carrier mothers can transmit variable and usually unpredictable amounts of mutant mtDNA to each of their offspring. A threshold effect exists, meaning that a certain level of heteroplasmy is required for symptoms to become manifest. This threshold varies between mutations and tissues and cannot be exactly defined for most individual mtDNA mutations due to lack of data. Consequently, PND is hampered by the often limited predictive value of the sample's mutation load for the expected phenotype. PGD can be an advantageous alternative (Sallevelt *et al.*, 2013), achieving pregnancies from embryos that are either mutation-free or harbor heteroplasmy levels safely below the phenotypic threshold, thereby circumventing arbitrary decisions regarding pregnancy termination. As said, for the majority of mtDNA mutations, insufficient data are available to determine a specific threshold. However, a systematic meta-analysis provided a generic threshold

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by showing a 95% or higher chance of being unaffected at (muscle) mutant level of 18% or less, irrespective of the mutation (Hellebrekers *et al.*, 2012), which can be used as input for counseling. Worldwide only a handful of centers perform PGD for mtDNA mutations (Steffann *et al.*, 2006; Thorburn *et al.*, 2009; Irving *et al.*, 2012; Treff *et al.*, 2012; Vandewoestyne *et al.*, 2012; Sallevelt *et al.*, 2013).

Here, we report the first PGD for the m.14487 T>C mutation in the *MT-ND6* gene causative of Leigh syndrome. For this indication, two blastomeres were analyzed, because data on whether a single blastomere is representative of the m.14487 T>C mutation for the entire embryo were lacking, and for other mtDNA mutations single blastomeres are known to incidentally harbor deviating heteroplasmy levels (Sallevelt *et al.*, 2013). The generic threshold level for transfer of 18% was applied (Hellebrekers *et al.*, 2012) as for the m.14487 T>C mutation relatively limited data were available. The mutation has been reported with details on phenotype and mutation load in a total of 22 affected individuals and 14 unaffected family members (Lebon *et al.*, 2003; Solano *et al.*, 2003; Ugalde *et al.*, 2003; Bugiani *et al.*, 2004; Raspall-Chaure *et al.*, 2004; Esteitie *et al.*, 2005; Malfatti *et al.*, 2007; Naess *et al.*, 2009; Wang *et al.*, 2009; Dermaut *et al.*, 2010; Leshinsky-Silver *et al.*, 2011; Spyropoulos *et al.*, 2013; Tarnopolsky *et al.*, 2013; Eckenweiler *et al.*, 2015). It appears that for the m.14487 T>C mutation rather high heteroplasmy levels are needed for a clinically significant phenotype; 13 of 15 (87%) patients with established muscle mutation loads harbored ≥80% heteroplasmy and 10 of 15 ≥93%, respectively. The lowest mutation load in muscle detected in an affected individual was 50%. These findings suggest that a higher threshold than the generic threshold of 18% might be applied for the m.14487 T>C mutation. Mutation loads up to >80% have been reported in some of the unaffected family members (e.g. Wang *et al.*, 2009). However, in none of the unaffected carriers, mutation load was determined in muscle, making it impossible to reliably evaluate the heteroplasmy range in asymptomatic carriers. Blood mutation levels of the m.14487 T>C are not representative since these were often significantly lower compared to levels in other tissues in patients as well, even in young and/or seriously affected patients (Dermaut *et al.*, 2010; Leshinsky-Silver *et al.*, 2011; Spyropoulos *et al.*, 2013) and have indeed been shown to decrease over time in one child (Solano *et al.*, 2003).

Case report

Case description

A 39-year-old female carrier of the m.14487 T>C mutation in the *MT-ND6* gene (Bugiani *et al.*, 2004) was referred from Italy. She and her husband requested PGD to fulfill their desire for a healthy child. Their son, 8 years of age at that time, had Leigh syndrome with m.14487 T>C mutation loads of >95% in muscle, fibroblasts and lymphocytes, respectively. Growth delay was noted at 6 months of age. Subsequently, stagnation of psychomotor development with hypotonia became apparent, and his brain MRI showed lesions typical of Leigh syndrome. He had feeding difficulties and recurrent aspiration pneumonias, necessitating tube feeding. At age 1, a cardiac arrest due to respiratory failure resulted in an intensive care admission. He was discharged with a tracheostomy. Years of intensive home caring followed. Eventually, the boy died at age 11 of a respiratory infection. His asymptomatic mother carried the mutation at heteroplasmy levels of ~32% in lymphocytes, ~65% in urine and ~55% in muscle. The mutation was also present in the mother's sister and mother, but not in her maternal grandmother and uncle (Fig. 1).

The couple was informed of the PGD procedure as customary (Sallevelt *et al.*, 2013) and in this case they were specifically made aware that PGD for the m.14487 T>C had never been carried out. Written consent, which included the transfer threshold of 18%, was obtained. By the time the PCR protocol had been developed and validated, the female was 40 years old. At that time, her basal FSH and 17-beta-estradiol levels were 5.6 U/l and 0.15 nmol/l, respectively, and antral follicle counts were 12 and 8 at cycle Day 4, pointing to a good ovarian reserve despite her age. There were no contraindications for the IVF procedure concerning her own health, a cardiologic evaluation revealed no abnormalities.

Methods

m.14487 T>C mutation levels in the female, her son and other family members were quantified by fluorescent last-cycle restriction fragment length polymorphism analysis, using a reverse modified primer and the TspRI enzyme (Department of Molecular Neurogenetics, National Institute of Neurology 'C.Besta', Milano, Italy) (Bugiani *et al.*, 2004).

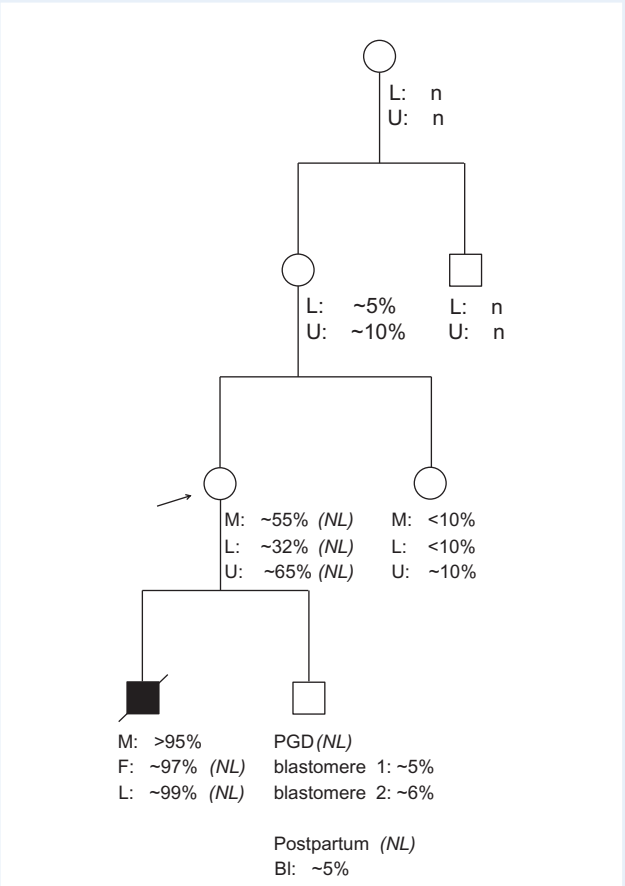


Figure 1 Pedigree. M, muscle; F, fibroblasts; L, lymphocytes; U, urine; BM, buccal mucosa; n, normal (mutation not detected); (NL), genetic analysis performed in the Netherlands, Maastricht University Medical Centre (the other analyses have been carried out in Italy, Milano National Institute of Neurology 'C.Besta').

ICSI and PGD procedures (including analysis of oocytes and remaining embryos) were performed as previously described, as were PCR and single-cell quantification assay (Sallevelt *et al.*, 2013) only with the use of other primers. Mismatch primer ND6_340 R ATAGTTTTTTT AATTTATTTAGGGGCA and primer ND6_199 F (6-FAM)-ATACT CTTTCACCCACAGCAC were used to detect the m.14487 T>C mutation by creating a TspRI Restriction site when the mutation was present. The primers SpikeTspRI-F VIC-AGC TGG TAA AAA GAG GCC TAA C and SpikeTspRI-R GCG CCG AAT AAT AGG TAT AGT G were used to generate an amplicon containing a TspRI site. This VIC labeled amplicon was used to spike the m.14487 T>C digestion samples to control for complete digestion with the restriction enzyme (TspRI) in order to identify and quantify the m.14487 T>C mutation. A calibration curve showing the linearity of the amplification and a Shewhart plot demonstrating the reproducibility are included as Supplementary Figures S1 and S2.

Cleavage stage embryos were biopsied at Day 3 postfertilization (which is typically the 8-cell stage), and two blastomeres were analyzed. In the individual blastomeres of the non-transferred and non-frozen embryos, as well as in unfertilized oocytes, m.14487 T>C heteroplasmy levels were determined in order to obtain further information about interblastomere mutation load variation of this mutation. If embryos could not entirely be dissected in separate blastomeres due to strong intercellular interactions in the morula stage, mean mutation loads of multiple blastomeres were established.

Ethical approval

The study was approved by the Medical Research Ethics Committee azM/UM (reference number 03-199).

Patient consent

Patient consent for publication was obtained.

Results

Mutation loads identified during the PGD treatment cycles are visualized in Fig. 2 and annotated in Supplementary Table S1.

The first cycle yielded 12 oocytes, 11 were successfully fertilized and 8 developed into 2-pronuclear (2PN) embryos eligible for 2-blastomere biopsy at Day 3 postfertilization. Mutation range varied from ~50 to ~98%, so unfortunately, all embryos showed m.14487 T>C heteroplasmy levels above the threshold and none could therefore be transferred. Also the unfertilized oocyte showed a high mutation load. Of the three remaining fertilized oocytes, two were degenerated and could not be analyzed, the other one, a single-cell apronuclear (0PN) zygote did not give a result upon testing. In the second cycle, 18 oocytes were retrieved and fertilized, resulting in 15 2PN embryos and three 3PN embryos with mutant load ranging from ~6 to ~100%. Thirteen 2PN embryos consisted of ≥7 blastomeres enabling 2-blastomere biopsy. Of the two embryos consisting of five blastomeres, a single cell was biopsied. Two embryos showed

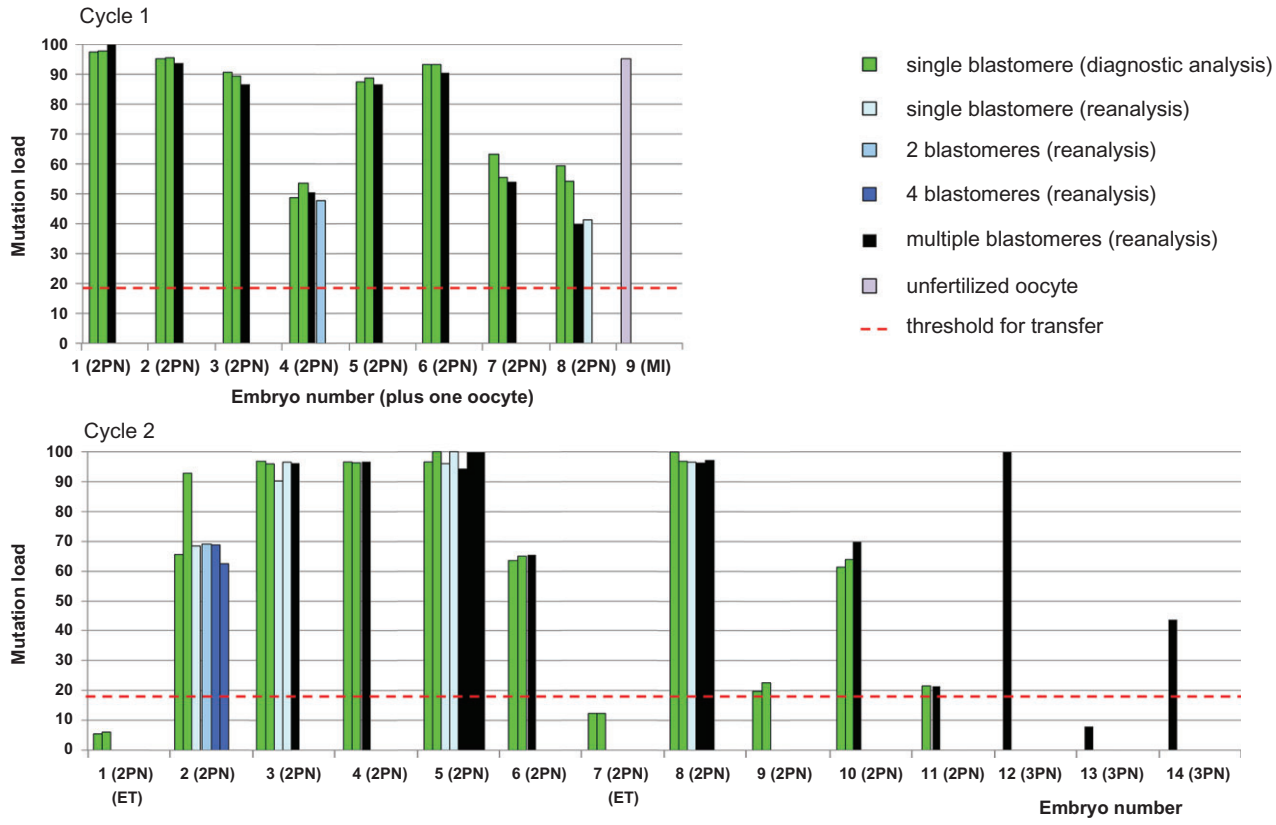


Figure 2 m.14487 T>C PGD treatment cycles. MI, metaphase I; PN, pronuclear stage; ET, embryo transfer.

heteroplasmy levels below the transfer threshold: mutation loads following 2-blastomere biopsy were 5.4 and 6.1% in a 12-cell embryo (embryo 1 of cycle 2) and 12.2 and 12.3% in a 13-cell embryo (embryo 7 of cycle 2), respectively. These two embryos were transferred. In three 8-cell embryos and one of the 5-cell embryos, the result (in all four above the transfer threshold) was inconclusive because of a positive blank. These are not included in Fig. 2/Supplementary Table S1.

Maximum differences in mutation loads between individual blastomeres were between 0 and 5% [0.1–3.6%] in 12 embryos, between 5 and 10% [5.8–9.1%] in five embryos and 19.4% (embryo 8 of cycle 1) and 30.6% (embryo 2 of cycle 2), in the remaining two embryos.

The transfer of the two embryos with mutation loads below 18% resulted in a viable single pregnancy, from which a healthy son was born at mother's age 42. PND to confirm the PGD diagnosis was offered but declined by the couple. In the boy's cord blood the m.14487 T>C mutation was shown with ~5% heteroplasmy, which was in accordance with the mutation load of one of the transferred embryos. He is now 7 months old, is healthy and is developing completely normal.

Discussion

We performed a successful PGD treatment in a female m.14487 T>C carrier, resulting in the birth of a healthy boy. This is the first PGD reported for this mutation.

mtDNA segregation and genetic bottleneck

Our data illustrate the unpredictable segregation of mtDNA mutations from a mother to her offspring. The female carrier is healthy and has moderate mutation loads in her tissues, whereas a significant proportion of her offspring (i.e. the affected child and a subset of the embryos) harbored much higher heteroplasmy levels. The possibility of this to occur is inherent to mtDNA inheritance. The number of mtDNA molecules is believed to drastically reduce in the female germline upon transmission, referred to as a 'genetic bottleneck' (e.g. Cree *et al.*, 2008). As a result, each offspring receives mtDNA that has been formed out of a small (mostly random) sample from the total maternal mtDNA pool. Accordingly, the resulting mutation levels in each offspring may vary widely, both compared to each other and to the mother.

Based on the overrepresentation of embryos with >90% mutant load produced by this carrier one may hypothesize the occurrence of selection in favor of the mutation upon transmission, which might at first sight seem corroborated by a number of cases whereby the mother has low mutation load and the child high (Lebon *et al.*, 2003; Solano *et al.*, 2003; Esteitie *et al.*, 2005). However, such shifts can occur due to the bottleneck (Otten/Sallevelt *et al.*, unpublished results) and the reported data mostly represent a single transmission event (one child). Furthermore, there are also reports of more variable segregation patterns, even including the apparent decrease in mutation load upon generations (Wang *et al.*, 2009; Dermaut *et al.*, 2010). Thus, current available data are insufficient, both in number and comparability (different tissues tested), to draw conclusions. At the cellular level, there is so far no evidence for a selection event for this mutation and its inheritance seems therefore most likely determined by random genetic drift (Otten/Sallevelt *et al.*, unpublished results).

Representativeness of blastomere mutation load

Critical for the reliability of a PGD treatment for mtDNA mutations is the representativeness of the mutation load in the biopsied blastomere for the entire embryo. Blastomere mutation load is generally fairly constant within the embryo, although single outliers may occur (Steffann *et al.*, 2006; Sallevelt *et al.*, 2013, 2015). These outliers are probably the result of chance, as the high amount of mtDNA molecules present in the oocyte (50 000–550 000 (Fragouli *et al.*, 2015)) is presumed to be partitioned randomly over the daughter cells (blastomeres) upon cleavage of the early embryo.

We found differences of <10% in m.14487 T>C mutation load between cells of the same embryo in the majority (90%; 17/19), whereas in two cleavage stage embryos larger differences of ~20 and ~30%, respectively, were present. Despite the considerable heteroplasmy variation in these two embryos, neither were at risk of misclassification, because all blastomeres had mutant levels above the transfer threshold. If the one deviating blastomere would have had a mutant level below the threshold whereas the other blastomeres did not, single-blastomere biopsy would have risked the unjust transfer of an embryo. However, in a recent study of another mtDNA mutation (m.3243 A>G), we have shown that this risk is very low (0.34%), justifying a single blastomere protocol with advantageous effects on embryo viability, and we conclude that this is applicable to other mtDNA mutations as well (Sallevelt *et al.*, unpublished results). Therefore, henceforth PGD for the m.14487 T>C mutation can be carried out on a single blastomere instead of two.

In comparison to trophectoderm (TE) biopsy, blastomere biopsy is more deleterious to the embryo (Scott *et al.*, 2013). However, until it is shown that TE biopsy gives a true representation of the heteroplasmy percentage in the inner cell mass, blastomere biopsy remains preferred for mtDNA mutations.

In conclusion, the current report illustrates the feasibility of PGD for mtDNA mutation carriers who wish to conceive a biologically own child, whereby the blastomere mutation load appears representative for postnatal heteroplasmy levels. PGD is clearly preferred compared with PND for carriers such as the female described here, who produce many embryos harboring high mutation loads and a few with low mutation levels which can be conveniently selected this way. This may take more than one PGD treatment or more oocytes per cycle, but likely prevents a large number of abnormal (and/or uninterpretable) PND results. Furthermore, this case confirms earlier reports that the majority of mtDNA mutation carriers can benefit from PGD to prevent the transmission of mtDNA disease to their offspring (Hellebrekers *et al.*, 2012; Steffann *et al.*, 2014; Sallevelt et al., 2015), and only a small subset may need the currently still experimental and more controversial nuclear transfer techniques (Tachibana *et al.*, 2013; Hyslop *et al.*, 2016; Zhang *et al.*, 2016).

Supplementary data

Supplementary data are available at *Human Reproduction* online.

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Authors' roles

S.C.E.H.S.: Contributions to conception and design, acquisition, analysis and interpretation of data. Drafting the article. Final approval of the version to be published. J.C.F.M.D., D.M.E.I.H., A.D.C.P. and E.C.: Contributions to conception and design, acquisition, analysis and interpretation of data. Critical review of the article. Final approval of the version to be published. M.D.: Contributions to acquisition, analysis and interpretation of data. Critical review of the article. Final approval of the version to be published. R.J.T.v.G., L.G., M.C.M., and M.Z.: Contributions to acquisition of data. Critical review of the article. Final approval of the version to be published. J.P.M.G.: Contributions to conception and design. Critical review of the article. Final approval of the version to be published. H.J.M.S. and C.E.M.d.D.S.: Contributions to conception and design, acquisition, analysis and interpretation of data. Critical review and supervising the drafting of the article. Final approval of the version to be published.

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Conflict of interest

None declared.

References

Bugiani M, Invernizzi F, Alberio S, Briem E, Lamantea E, Carrara F, Moroni I, Farina L, Spada M, Donati MA *et al*. Clinical and molecular findings in children with complex I deficiency. *Biochim Biophys Acta* 2004; **1659**:136–147.

Cree LM, Samuels DC, de Sousa Lopes SC, Rajasimha HK, Wonnapinij P, Mann JR, Dahl HH, Chinnery PF. A reduction of mitochondrial DNA molecules during embryogenesis explains the rapid segregation of genotypes. *Nat Genet* 2008;**40**:249–254.

Dermaut B, Seneca S, Dom L, Smets K, Ceulemans L, Smet J, De Paepe B, Tousseyn S, Weckhuysen S, Gewillig M *et al*. Progressive myoclonic epilepsy as an adult-onset manifestation of Leigh syndrome due to m. 14487 T>C. *J Neurol Neurosurg Psychiatry* 2010;**81**:90–93.

Eckenweiler M, Catarino CB, Gallenmueller C, Klopstock T, Lagreze WA, Korinthenberg R, Kirschner J. Mitochondrial DNA mutation 14487 T>C manifesting as Leber's hereditary optic neuropathy. *J Neurol* 2015;**262**: 2776–2779.

Esteitie N, Hinttala R, Wibom R, Nilsson H, Hance N, Naess K, Tear-Fahnehjelm K, von Dobeln U, Majamaa K, Larsson NG. Secondary metabolic effects in complex I deficiency. *Ann Neurol* 2005;**58**:544–552.

Fragouli E, Spath K, Alfarawati S, Kaper F, Craig A, Michel CE, Kokocinski F, Cohen J, Munne S, Wells D. Altered levels of mitochondrial DNA are associated with female age, aneuploidy, and provide an independent measure of embryonic implantation potential. *PLoS Genet* 2015;**11**:e1005241.

Gerards M, Sallevelt SC, Smeets HJ. Leigh syndrome: resolving the clinical and genetic heterogeneity paves the way for treatment options. *Mol Genet Metab* 2016;**117**:300–312.

Hellebrekers DM, Wolfe R, Hendrickx AT, de Coo IF, de Die CE, Geraedts JP, Chinnery PF, Smeets HJ. PGD and heteroplasmic mitochondrial DNA point mutations: a systematic review estimating the chance of healthy offspring. *Hum Reprod Update* 2012;**18**:341–349.

Hyslop LA, Blakeley P, Craven L, Richardson J, Fogarty NM, Fragouli E, Lamb M, Wamaitha SE, Prathalingam N, Zhang Q *et al*. Towards clinical application of pronuclear transfer to prevent mitochondrial DNA disease. *Nature* 2016;**534**:383–386.

Irving L, Craven L, Alston C, Byerley S, Taylor R, Herbert M, Turnbull D. Determination of mutation loads in preimplantation embryos to assess the effectiveness of preimplantation genetic diagnosis (PGD) for mitochondrial DNA (mtDNA) inherited disorders. *Reprod Biomed Online* 2012;**24**:S20.

Johnston IG, Burgstaller JP, Havlicek V, Kolbe T, Rulicke T, Brem G, Poulton J, Jones NS. Stochastic modelling, Bayesian inference, and new in vivo measurements elucidate the debated mtDNA bottleneck mechanism. *eLife* 2015;**4**:e07464.

Lebon S, Chol M, Benit P, Mugnier C, Chretien D, Giurgea I, Kern I, Girardin E, Hertz-Pannier L, de Lonlay P *et al*. Recurrent de novo mitochondrial DNA mutations in respiratory chain deficiency. *J Med Genet* 2003;**40**:896–899.

Leshinsky-Silver E, Shuvalov R, Inbar S, Cohen S, Lev D, Lerman-Sagie T. Juvenile Leigh syndrome, optic atrophy, ataxia, dystonia, and epilepsy due to T14487C mutation in the mtDNA-ND6 gene: a mitochondrial syndrome presenting from birth to adolescence. *J Child Neurol* 2011;**26**: 476–481.

Malfatti E, Bugiani M, Invernizzi F, de Souza CF, Farina L, Carrara F, Lamantea E, Antozzi C, Confalonieri P, Sanseverino MT *et al*. Novel mutations of ND genes in complex I deficiency associated with mitochondrial encephalopathy. *Brain* 2007;**130**:1894–1904.

Naess K, Freyer C, Bruhn H, Wibom R, Malm G, Nennesmo I, von Dobeln U, Larsson NG. MtDNA mutations are a common cause of severe disease phenotypes in children with Leigh syndrome. *Biochim Biophys Acta* 2009;**1787**:484–490.

Raspall-Chaure M, Solano A, Vazquez E, Macaya-Ruiz A, del Toro-Riera M, Cabezuelo-Briones A, Montoya J, Andreu A, Roig-Quilis M. A patient with bilateral lesion in the striatum and slowly progressive dystonia secondary to T14487C mutation in the ND6 gene of complex I of the mitochondrial respiratory chain. *Rev Neurol* 2004;**39**:1129–1132.

Sallevelt SC, Dreesen JC, Drusedau M, Spierts S, Coonen E, van Tienen FH, van Golde RJ, de Coo IF, Geraedts JP, de Die-Smulders CE *et al*. Preimplantation genetic diagnosis in mitochondrial DNA disorders: challenge and success. *J Med Genet* 2013;**50**:125–132.

Sallevelt SCEH JCFMD, de Coo IFM, de Die-Smulders CEM, Smeets HJM. Selecting the right embryo in mitochondrial disorders. In: Sills ES (ed). *Screening the Single Euploid Embryo – Molecular Genetics in Reproductive Medicine*. Switzerland: Springer International Publishing, 2015, 213–246.

Scott RT Jr., Upham KM, Forman EJ, Zhao T, Treff NR. Cleavage-stage biopsy significantly impairs human embryonic implantation potential while blastocyst biopsy does not: a randomized and paired clinical trial. *Fertil Steril* 2013;**100**:624–630.

Solano A, Roig M, Vives-Bauza C, Hernandez-Pena J, Garcia-Arumi E, Playan A, Lopez-Perez MJ, Andreu AL, Montoya J. Bilateral striatal necrosis associated with a novel mutation in the mitochondrial ND6 gene. *Ann Neurol* 2003;**54**:527–530.

Spyropoulos A, Manford M, Horvath R, Alston CL, Yu-Wai-Man P, He L, Taylor RW, Chinnery PF. Near-identical segregation of mtDNA heteroplasmy in blood, muscle, urinary epithelium, and hair follicles in twins

with optic atrophy, ptosis, and intractable epilepsy. *JAMA Neurol* 2013; **70**:1552–1555.

Steffann J, Frydman N, Gigarel N, Burlet P, Ray PF, Fanchin R, Feyereisen E, Kerbrat V, Tachdjian G, Bonnefont JP *et al*. Analysis of mtDNA variant segregation during early human embryonic development: a tool for successful NARP preimplantation diagnosis. *J Med Genet* 2006;**43**:244–247.

Steffann J, Gigarel N, Samuels DC, Monnot S, Borghese R, Hesters L, Frydman N, Burlet P, Frydman R, Benachi A *et al*. Data from artificial models of mitochondrial DNA disorders are not always applicable to humans. *Cell Rep* 2014;**7**:933–934.

Tachibana M, Amato P, Sparman M, Woodward J, Sanchis DM, Ma H, Gutierrez NM, Tippner-Hedges R, Kang E, Lee HS *et al*. Towards germline gene therapy of inherited mitochondrial diseases. *Nature* 2013;**493**: 627–631.

Tarnopolsky M, Meaney B, Robinson B, Sheldon K, Boles RG. Severe infantile leigh syndrome associated with a rare mitochondrial ND6 mutation, m.14487 T>C. *Am J Med Genet A* 2013;**161a**:2020–2023.

Thorburn DR, Wilton L, Stock-Myer S. Healthy baby girl born following pre-implantation genetic diagnosis for mitochondrial DNA m.8993 T>G mutation. *Mol Genet Metab* 2009;**98**:5–6.

Treff NR, Campos J, Tao X, Levy B, Ferry KM, Scott RT Jr. Blastocyst preimplantation genetic diagnosis (PGD) of a mitochondrial DNA disorder. *Fertil Steril* 2012;**98**:1236–1240.

Ugalde C, Triepels RH, Coenen MJ, van den Heuvel LP, Smeets R, Uusimaa J, Briones P, Campistol J, Majamaa K, Smeitink JA *et al*. Impaired complex I assembly in a Leigh syndrome patient with a novel missense mutation in the ND6 gene. *Ann Neurol* 2003;**54**:665–669.

Vandewoestyne M, Heindryckx B, De Gheselle S, Lepez T, Neupane J, Gerris J, Van Coster R, De Sutter P, Deforce D. Poor correlation between polar bodies and blastomere mutation load in a patient with m. 3243 A>G tRNALeu(UUR) point mutation. *Mitochondrion* 2012;**12**: 477–479.

Wang J, Brautbar A, Chan AK, Dzwiniel T, Li FY, Waters PJ, Graham BH, Wong LJ. Two mtDNA mutations 14487 T>C (M63V, ND6) and 12297 T>C (tRNA Leu) in a Leigh syndrome family. *Mol Genet Metab* 2009;**96**:59–65.

Zhang J, Liu H, Luo S, Chavez-Badiola A, Liu Z, Yang M, Munne S, Konstantinidis M, Wells D, Huang T. First live birth using human oocytes reconstituted by spindle nuclear transfer for mitochondrial DNA mutation causing Leigh syndrome. *Fertil Steril* 2016;**106**: e375–e376.